The Nyquist Stability Criterion

Control Engineering (Reguleringsteknik)

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Agenda

Introduction

Frequency Response

Bode Plots

The Nyquist Stability Criterion

Stability Margins

Summary

Introduction
Curriculum for Reguleringsteknik (REG)

Matematiske og grafiske metoder til syntese af lineære tidsinvariante systemer:[1]

- diskret og kontinuert tilstandsbeskrivelse
- analyse i tid og frekvens
- stabilitet, reguleringshastighed, følsomhed og fejl
- digitale PI, PID, LEAD og LAG regulatorer (serieregulatorer)
- tilstandsregulering, pole-placement og tilstands-estimering (observer)
- optimal regulering (least squares) og optimal tilstands-estimation (Kalman-filter)

Færdigheder:
Efter gennemførelse af kurset kan den succesfulde studerende:

- kunne analysere, dimensionere og implementere såvel kontinuert som tidsdiskret regulering af lineære tidsinvariante og stokastiske systemer

Kompetencer:
Efter gennemførelse af kurset kan den succesfulde studerende:

- anvende og implementere klassiske og moderne reguleringsteknikker for at kunne styre og regulere en robot hurtig og præcist

[1] Based on https://fagbesk.sam.sdu.dk/?fag_id=39673

Introduction
Course Overview

The twelve lectures of the course are

- **Lecture 1**: Introduction to Linear Time-Invariant Systems
- **Lecture 2**: Stability and Performance Analysis
- **Lecture 3**: Introduction to Control
- **Lecture 4**: Design of PID Controllers
- **Lecture 5**: Root Locus
- **Lecture 6**: The Nyquist Plot
- **Lecture 7**: Dynamic Compensators and Stability Margins
- **Lecture 8**: Implementation
- **Lecture 9**: State Feedback
- **Lecture 10**: Observer Design
- **Lecture 11**: Optimal Control (Linear Quadratic Control)
- **Lecture 12**: The Kalman Filter

Frequency Response

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Frequency Response

Definition

The ***frequency response*** of a system is the steady-state response of a system to a sinusoidal input.

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The output of a time-invariant system will have the same frequency as the sinusoidal input, but possibly with a different amplitude and phase.

Frequency Response

System Response (1)

Recall that the output of a time-invariant system is given by

$$y(t) = \int_{-\infty}^{\infty} h(\tau)u(t-\tau)d\tau$$

where y is the output, u is the input, and h is the impulse response of the system.

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$$u(t) = A\cos(\omega t)$$

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$$u(t) = A\cos(\omega t)$$

By Euler's formula ($e^{ia} = \cos a + i \sin a$) the output can be written as

$$u(t) = \frac{A}{2}\left(e^{j\omega t} + e^{-j\omega t}\right)$$

Frequency Response

System Response (2)

Motivated by the input given as exponentials, the system response to an input $u(t) = e^{st}$ with $s = \sigma + j\omega$ is computed below

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&= \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau e^{st} \\
&= H(s)e^{st}
\end{aligned}
$$

Frequency Response

System Response (3)

The expression for $y(t)$ is then given as follows for $u(t) = A\cos(\omega t)$

$$y(t) = \frac{A}{2}\left(H(j\omega)e^{j\omega t} + H(-j\omega)e^{-j\omega t}\right)$$

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The complex number $H(j\omega)$ is expressed in polar form as

$$H(j\omega) = M(\omega)e^{j\varphi(\omega)}$$

where $M(\omega)$ is the magnitude of $H(j\omega)$ and $\varphi(\omega)$ is the phase of $H(j\omega)$.

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This gives the following expression for the output

$$y(t) = \frac{A}{2}M(\omega)\left(e^{j(\varphi(\omega)+\omega t)} + e^{-j(\varphi(\omega)+\omega t)}\right)$$

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$$\begin{aligned} y(t) &= \frac{A}{2}M(\omega)\left(e^{j(\varphi(\omega)+\omega t)} + e^{-j(\varphi(\omega)+\omega t)}\right) \\ &= \frac{A}{2}M(\omega)\cos(\omega t + \varphi(\omega)) \end{aligned}$$

Frequency Response

System Response (4)

In conclusion, the system response subject to the input $u(t) = A\cos(\omega t)$ is

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This means that the output of the system is also a sinusoidal signal with the same frequency as the input, but possibly with an other amplitude and a phase shift.

Frequency Response
Example

Consider the first-order differential equation

$$\dot{y} + ky = u$$

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Frequency Response
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Consider the first-order differential equation

$$\dot{y} + ky = u$$

The Laplace transform of the differential equation leads to

$$H(s) = \frac{y(s)}{u(s)} = \frac{1}{s+k}$$

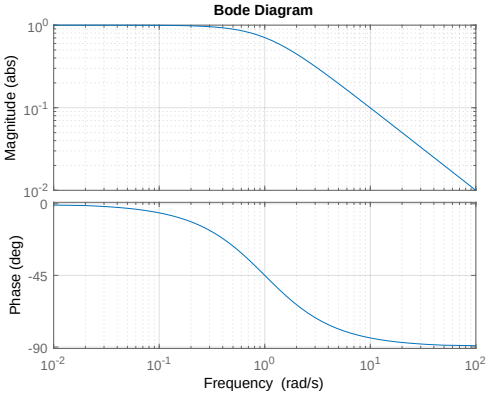
The ***frequency response*** of $H(s)$ is given by the magnitude and phase of $H(j\omega) = \frac{1}{j\omega+k}$

$$M = |H(j\omega)| = \frac{1}{\sqrt{\omega^2 + k^2}} \qquad \text{and} \qquad \varphi = -\tan^{-1}\left(\frac{\omega}{k}\right)$$

where M is the ***amplitude ratio*** and φ is the ***phase***.

Frequency Response

Example



Frequency Response

Second-Order System (1)

The transfer function of a second-order system can be written as

$$G(s) = \frac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

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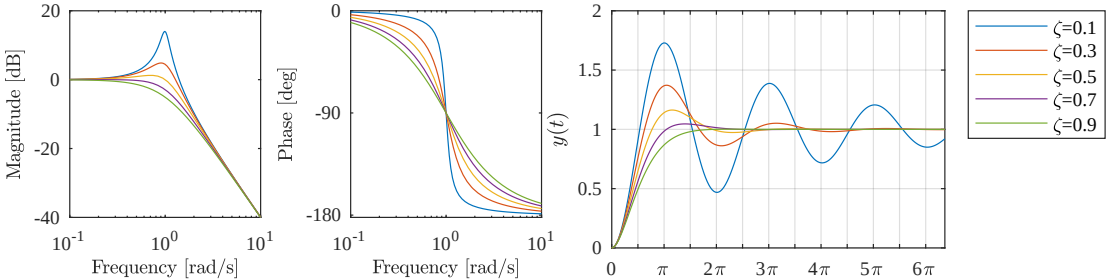

Frequency Response
Terminology

The ***bandwidth*** of a closed-loop system $T(s)$ is defined to be the maximum frequency at which the output y of a system will track a sinusoidal input r in a satisfactory manner (output attenuated to $1/\sqrt{2}$ times the input).
Formally, the bandwidth ω_{BW} of $T(s)$ is the maximal frequency such that

$$T(j\omega) \geq \frac{1}{\sqrt{2}} \qquad \forall \omega \in [0, \omega_{BW}]$$

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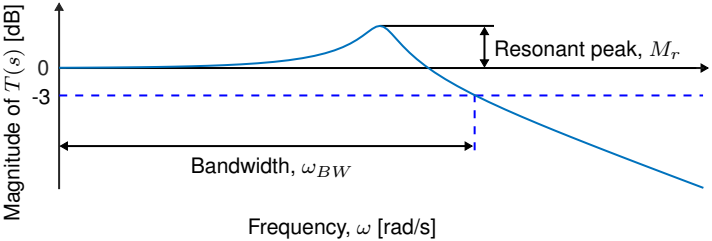
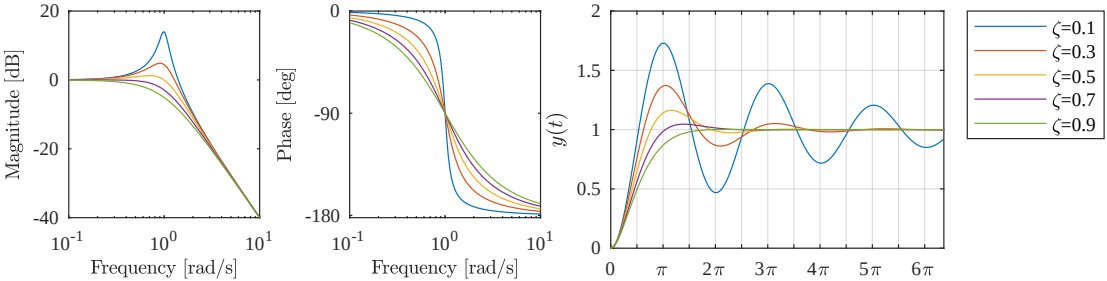


Frequency Response

Second-Order System (2)

For a second-order system

- **Bandwidth**: The bandwidth $\omega_{BW} \approx \omega_n$ (the equality holds for $\zeta = 0.7$).
- **Resonant peak**: The resonant peak is $M_p = \frac{1}{2\zeta}$ for $\zeta < 0.5$.



Bode Plots

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Bode Plots

Introduction

A transfer function $G(s)$ evaluated at $s = j\omega$ can be be factored in terms of poles and zeros, and written as

$$G(j\omega) = \frac{r_1 e^{j\theta_1} r_2 e^{j\theta_2}}{r_3 e^{j\theta_3} r_4 e^{j\theta_4} r_5 e^{j\theta_5}} = \frac{r_1 r_2}{r_3 r_4 r_5} e^{j(\theta_1+\theta_2-\theta_3-\theta_4-\theta_5)}$$

Bode Plots

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The the magnitude of $G(j\omega)$ is

$$|G(j\omega)| = \frac{r_1 r_2}{r_3 r_4 r_5}$$

which leads to

$$\log_{10} |G(j\omega)| = \log_{10} r_1 + \log_{10} r_2 - \log_{10} r_3 - \log_{10} r_4 - \log_{10} r_5$$

Bode Plots
Advantages

Advantages of working with Bode plots

1. Dynamic compensator design can be based entirely on the Bode plots. (We will do this in the next lecture)
2. Bode plots can be determined experimentally.
3. The Bode plot of a system defined by a series connection is given by the addition of the Bode plots of the individual systems.
4. The use og log scale permits a much wider range of frequencies to be displayed on a single plot than is possible with linear scales.

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Example: Consider the series connection of two systems

$$G_1(s) = \frac{1}{(s+1)^2} \qquad G_2(s) = s + 100$$

Bode Plots

Bode Form of Transfer Function

In the following, we consider the open-loop transfer function

$$KG(s) = K\frac{(s-z_1)(s-z_2)\cdots}{(s-p_1)(s-p_2)\cdots}$$

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$$KG(s) = K\frac{(s-z_1)(s-z_2)\cdots}{(s-p_1)(s-p_2)\cdots}$$

For convenience, the transfer function is evaluated at $s = j\omega$ and rewritten into *Bode form* as

$$KG(s) = K_0\frac{(j\omega\tau_1+1)(j\omega\tau_2+1)\cdots}{(j\omega\tau_a+1)(j\omega\tau_b+1)\cdots}$$

Bode Plots

Classes of Terms in Transfer Function

In the following, three classes of terms of the transfer function are analyzed

1. Class 1: $K_0(j\omega)^n$ for $n \in \mathbb{Z}$.
2. Class 2: $(j\omega\tau + 1)^{\pm 1}$.
3. Class 3: $\left((j\omega/\omega_n)^2 + 2\zeta(j\omega/\omega_n) + 1\right)^{\pm 1}$.

For the term $K_0(j\omega)^n$, we have

$$\log K_0|(j\omega)^n| = \log K_0 + n \log |j\omega|$$

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The phase is constant and given by

$$\varphi = n \cdot 90^\circ$$

Bode Plots

Sketching Bode Plot: Class 1 (2)

To sketch a Bode plot for $K_0(j\omega)^n$, we start at $\omega = 1$, then from

$$\log K_0|(j\omega)^n| = \log K_0 + n \log |j\omega|$$

it is seen that $\log K_0|(j)^n| = \log K_0$.

Bode Plots

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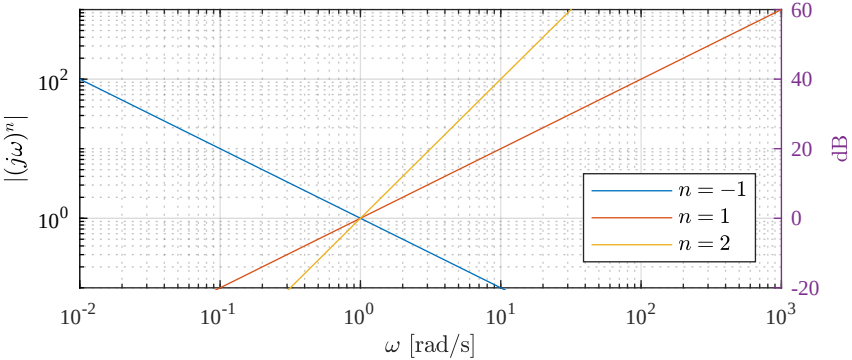

For the term $(j\omega\tau + 1)^{\pm 1}$, we study two asymptotes

- For $\omega\tau \ll 1$, $j\omega\tau + 1 \approx 1$.
- For $\omega\tau \gg 1$, $j\omega\tau + 1 \approx j\omega\tau$.

Bode Plots

Sketching Bode Plot: Class 2 (Magnitude)

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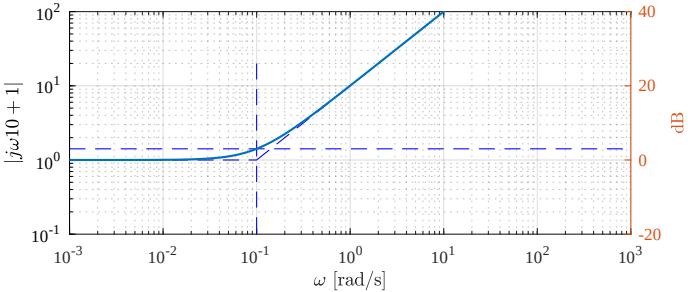

Bode Plots

Sketching Bode Plot: Class 2 (Phase)

For the phase of the term $(j\omega\tau + 1)^{\pm 1}$, we study two asymptotes

- For $\omega\tau \ll 1$, $\angle 1 = 0°$.
- For $\omega\tau \gg 1$, $\angle j\omega\tau = 90°$.

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In addition, at the break point $\omega = 1/\tau$, the angle is $\angle(1 + j) = 45°$.

Bode Plots

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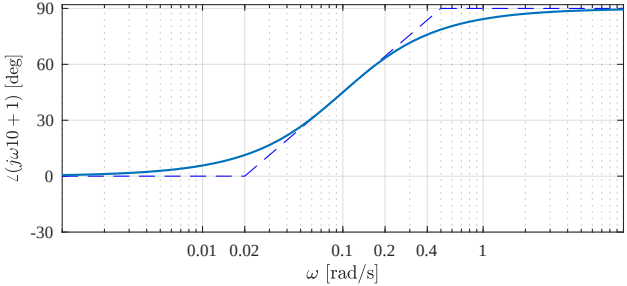

Bode Plots

Sketching Bode Plot: Class 3

For the term $\left((j\omega/\omega_n)^2 + 2\zeta(j\omega/\omega_n) + 1\right)^{\pm 1}$, we have

- For $\omega \ll \omega_n$, the amplitude is approximately 1.

Bode Plots

Sketching Bode Plot: Class 3

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- For $\omega \ll \omega_n$, the amplitude is approximately 1.

In addition, at the break point $\omega = \omega_n$, the magnitude is $|G(j\omega)| = 1/(2\zeta)$ and the phase is $\pm 90°$.

Bode Plots

Summary of Bode Plot Rules

1. Rewrite the considered transfer function into Bode form.
2. Determine the value of the $K_0(j\omega)^n$ term. Plot the low frequency magnitude asymptote through the point K_0 at $\omega = 1$ and with slope of $n\times$20/db per decade.
3. Complete the composite magnitude asymptotes by extending the low-frequency asymptote until the first frequency break point. Then change the slope according to the behavior at the break point, and continue the procedure for the remaining break points.
4. Sketch the approximate magnitude curve by increasing the asymptote value by a factor $\sqrt{2}$ at first-order numerator break and decreasing it by a factor $1/\sqrt{2}$ at denominator break points.

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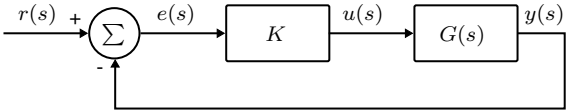
4. Sketch the approximate magnitude curve by increasing the asymptote value by a factor $\sqrt{2}$ at first-order numerator break and decreasing it by a factor $1/\sqrt{2}$ at denominator break points.
5. Plot the low-frequency asymptote of the phase $\phi = n \times 90°$.
6. Change the phase at the phase points, and correct the phase according to the slope at the phase point.

Bode Plots

Neutral Stability (1)

The stability of the closed-loop system shown below.



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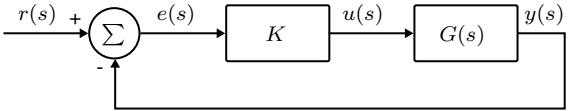

The stability of the system is determined from the frequency response of the *open-loop* transfer function $KG(s)$, which is usually known.

Recall the closed loop transfer function

$$T(s) = \frac{KG(s)}{1 + KG(s)}$$

Bode Plots

Neutral Stability (2)

The point that separates stable and unstable systems is associated to a closed-loop pole on the imaginary axis. At this point

$$|KG(s)| = 1 \quad \text{and} \quad \angle G(s) = 180^\circ$$

Bode Plots

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At the point of ***neutral stability***, the root locus conditions hold for $s = j\omega$, i.e.,

$$|KG(j\omega)| = 1 \quad \text{and} \quad \angle G(j\omega) = 180^\circ$$

Bode Plots

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The point that separates stable and unstable systems is associated to a closed-loop pole on the imaginary axis. At this point

$$|KG(s)| = 1 \quad \text{and} \quad \angle G(s) = 180^\circ$$

At the point of ***neutral stability***, the root locus conditions hold for $s = j\omega$, i.e.,

$$|KG(j\omega)| = 1 \quad \text{and} \quad \angle G(j\omega) = 180^\circ$$

This leads to the stability condition

$$|KG(j\omega)| < 1 \quad \text{and} \quad \angle G(j\omega) = -180^\circ$$

which holds for systems that become unstable, when the gain K increases.

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Consider the transfer function

$$H_1(s) = \frac{s^2 + 2s + 2}{(s+1)(s+5)}$$

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The Argument Principle

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$$H_1(s_0) = |v|e^{j\alpha}$$

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$$\alpha = \theta_1 + \theta_2 - \phi_1 - \phi_2$$

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The Nyquist Stability Criterion

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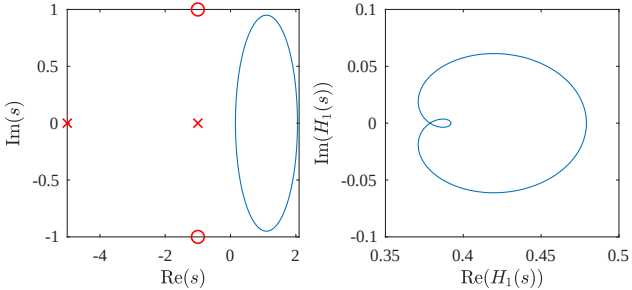
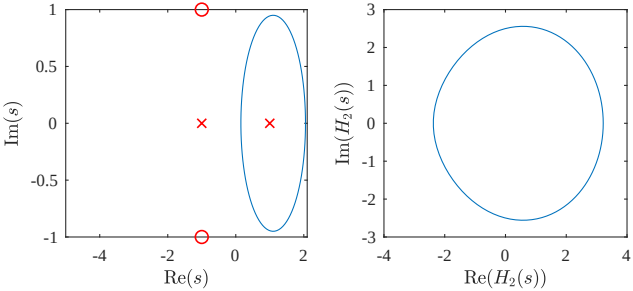


The Nyquist Stability Criterion

The Argument Principle

Consider the transfer function

$$H_2(s) = \frac{s^2 + 2s + 2}{(s+1)(s-1)}$$



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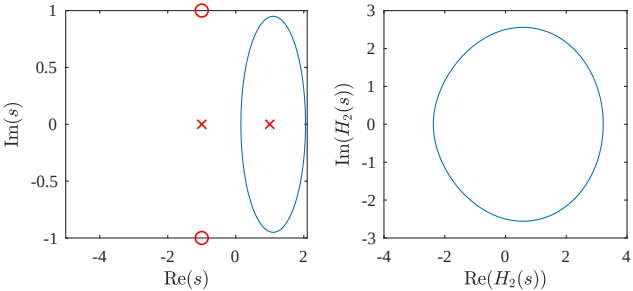

The contour encircles the origin in the counter clockwise direction.

Theorem. A contour map of a complex function will encircle the origin $Z - P$ times where Z is the number of zeros and P is the number of poles of the function inside the contour.

The Nyquist Stability Criterion

Stability

To verify the stability of a system, one needs to determine the number of closed-loop poles in the right half-plane. Thus, the Nyquist plot is a map of a contour that encircles the entire right-half plane.

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The number of closed-loop poles in the right half-plane equals the number of right half-plane zeros of

$$1 + KG(s)$$

To only work with the open-loop transfer function $KG(s)$, it suffices to count the number of encirclements of the point -1 when considering a contour that encircles the right half-plane for $KG(s)$.

The Nyquist Stability Criterion

Stability Criterion

Let N denote the number of clockwise encirclements of -1. Then the number of zeros in the right half plane Z (closed-loop poles) minus the number of open-loop poles in the right half-plane P is

$$N = Z - P$$

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Gain Margin: Definition

The ***gain margin*** is the factor by which the gain can be rained before a system becomes unstable.

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Recall that a system is ***neutrally stable*** for

$$|L(j\omega)| = 1 \quad \text{and} \quad \angle L(j\omega) = 180^\circ$$

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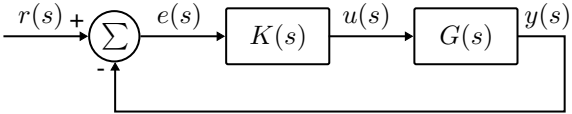


Stability Margins

Phase Margin: Definition

The ***phase margin*** is the amount by which the phase can be rained before a system becomes unstable (before it exceeds -$180°$).

A system is ***neutrally stable*** for

$$|L(j\omega)| = 1 \quad \text{and} \quad \angle L(j\omega) = 180°$$

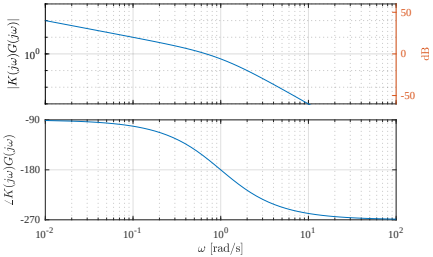
where $L(s) = K(s)G(s)$.

Stability Margins

Relation to Bode Plot

The stability margins can be determined from a Bode plot as (use `margin` in MATLAB)

- **Gain Margin**: Find the value of ω where $\angle K(j\omega)G(j\omega) = -180°$, and denote it by ω_{GM}. The gain margin (in dB) is
$$GM = -|K(j\omega_{GM})G(j\omega_{GM})|$$
- **Phase Margin**: Find the value of ω where $|K(j\omega)G(j\omega)| = 0$ dB, and denote it by ω_c (this frequency is called the ***crossover frequency***). The phase margin is
$$PM = \angle K(j\omega_c)G(j\omega_c) + 180°$$



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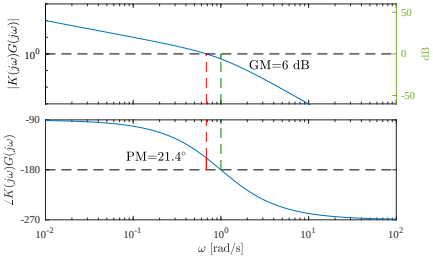
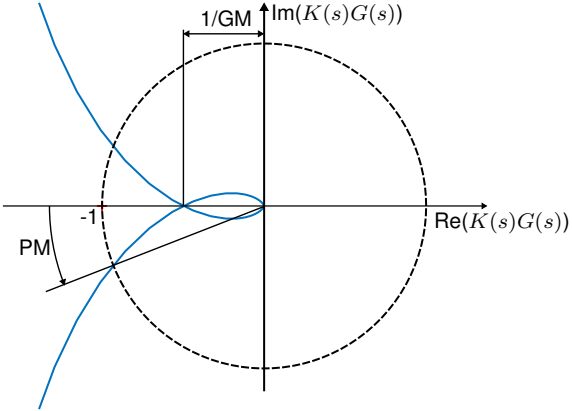


Stability Margins

Relation to Nyquist Plot

The stability margins can also be determined from a Nyquist plot, where the margins are determined by the closeness of the Nyquist plot to the point -1.



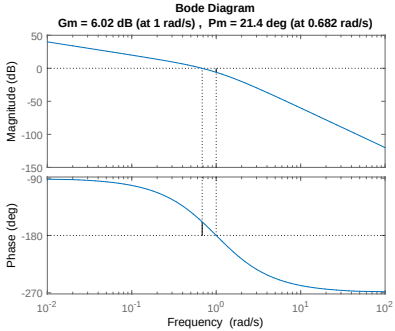
Stability Margins

Example: Interpretation of Margins

Consider the loop gain

$$L(s) = K(s)G(s) = \frac{1}{s(s+1)^2}$$

with the following Bode plot.



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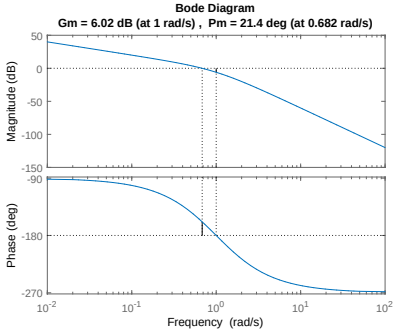

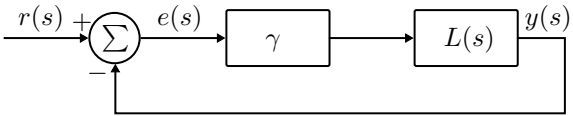

The gain γ can be increased to 2 before the system becomes unstable.

Stability Margins

Example: Parameter Change

Consider a loop gain

$$L(s) = \frac{1.3(s+2)}{s^3 + s^2 + bs + 1}$$

where b is some parameter.

Stability Margins

Example: Parameter Change

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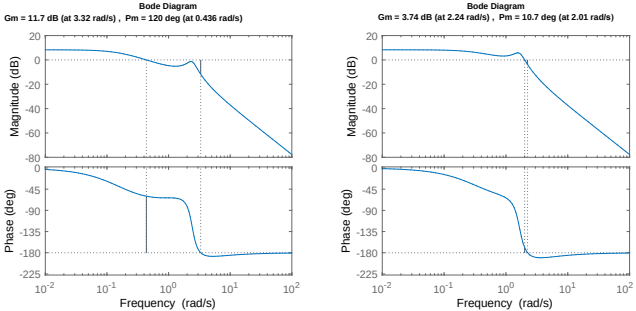
$$L(s) = \frac{1.3(s+2)}{s^3 + s^2 + bs + 1}$$

where b is some parameter. A parameter change from $b = 6$ (left) to $b = 3$ (right) changes the margins significantly.



Stability Margins

Example: Shortcoming of Phase and Gain Margins

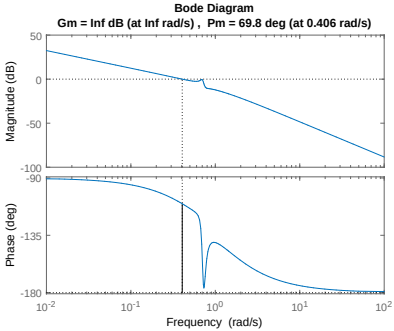
The phase and gain margins determine stability properties, when only the gain OR phase is changed.

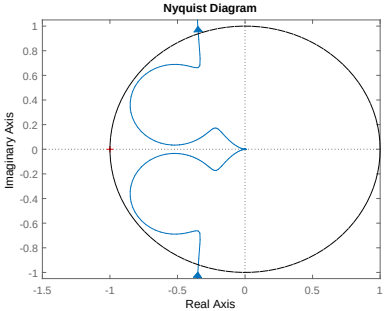
Stability Margins

Example: Shortcoming of Phase and Gain Margins

The phase and gain margins determine stability properties, when only the gain OR phase is changed.

If the gain and phase is changed simultaneously, issues can occur.



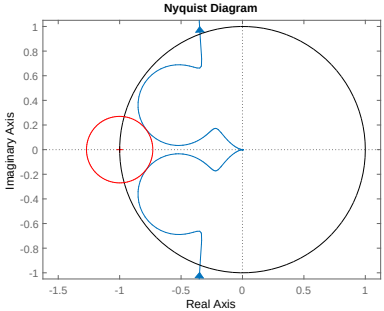


Stability Margins

Example: Shortcoming of Phase and Gain Margins

The phase and gain margins determine stability properties, when only the gain OR phase is changed.

This is the motivation for looking at the vector margin that gives the shortest distance between the Nyquist plot and -1.



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The stability margins determines the robustness of a control system based on the loop gain.

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$$GM = -|K(j\omega_{GM})G(j\omega_{GM})|$$

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